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Parameter Personalisasi

$N = 21$, $a = 2$, $b = 1$, $K = 4$, $\theta^0 = 65^\circ$, $\alpha = 0{,}05$

3) Definisi Himpunan Pelanggan

$A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\}$ -> Pelanggan dari medsos

$B = \{N+3, N+5, N+7, N+10, N+12, N+18\}$ -> pelanggan dari referal

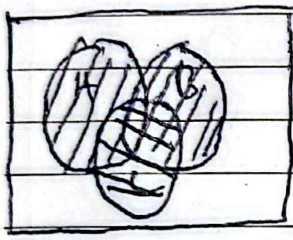
$C = \{N+1, N+7, N+10, N+11, N+17, N+20\}$ -> Pelanggan dari loyalty program

$A = \{21, 23, 25, 28, 31, 34, 36\}$

$B = \{24, 26, 28, 31, 33, 39\}$

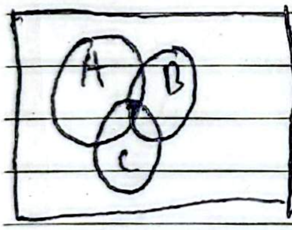
$C = \{22, 28, 31, 32, 38, 41\}$

$U = \{21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41\} = 21$



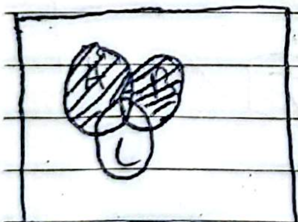
$A \cup B \cup C$ (semua himpunan digabung)

$= \{21, 22, 23, 24, 25, 26, 28, 31, 32, 33, 34, 36, 38, 39, 41\}$ = 15 anggota



$A \cap B \cap C$ (irisan dari ketiganya)

$= \{28, 31\}$ = 2 anggota

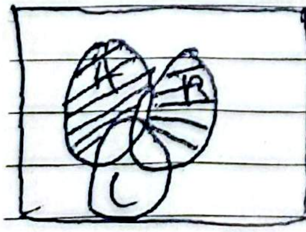


$(A \cup B) \cap C'$

$A \cup B = \{21, 23, 24, 25, 26, 28, 31, 33, 34, 36, 39\}$

$C' = \{21, 23, 24, 25, 26, 27, 29, 30, 33, 34, 35, 36, 37, 39, 40\}$

$(A \cup B) \cap C' = \{21, 23, 24, 25, 26, 33, 34, 36, 39\}$ = 9 anggota

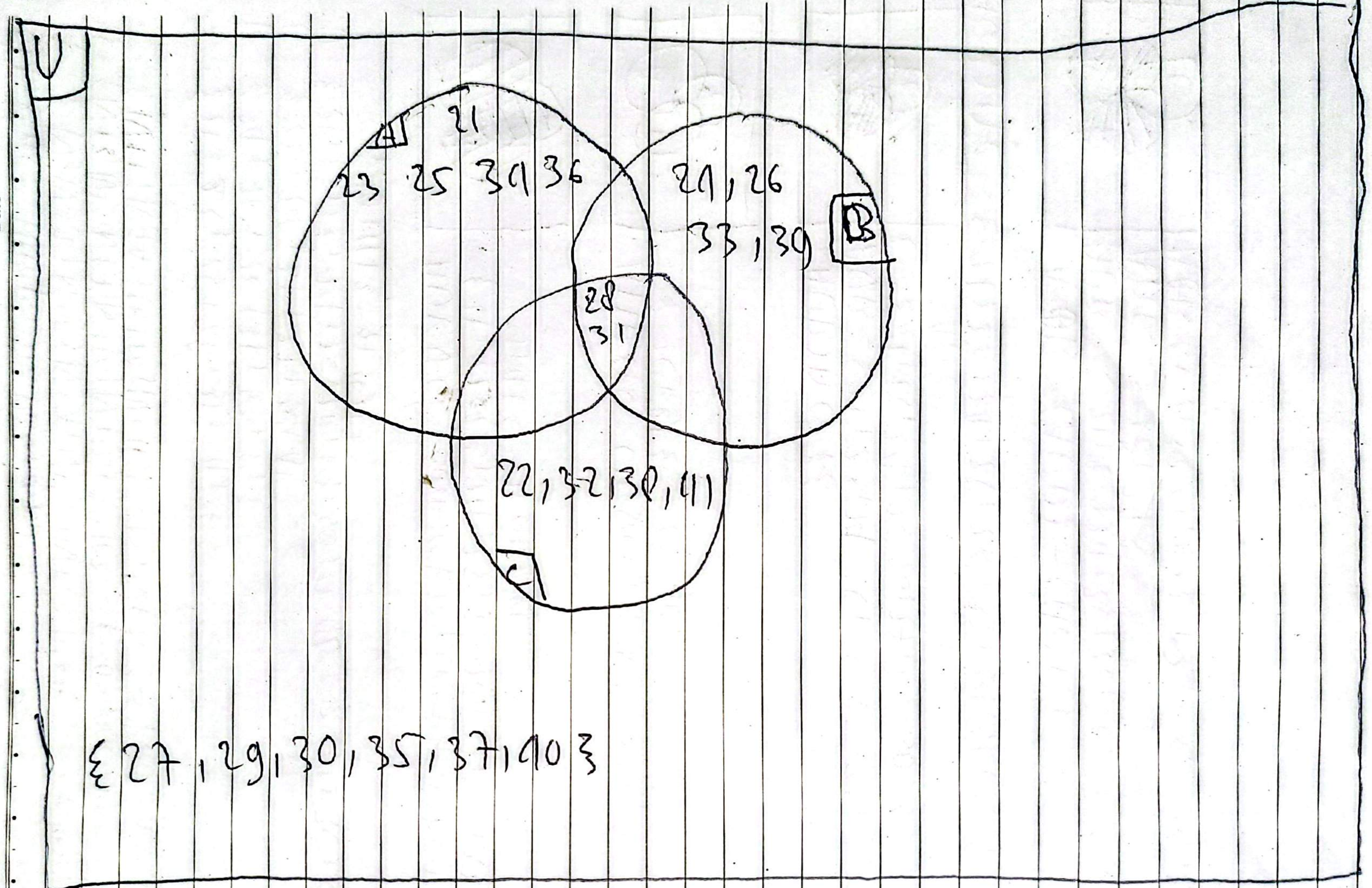


$A \oplus B = (A \cup B) - (A \cap B)$

$A \cup B = \{21, 23, 24, 25, 26, 28, 31, 33, 34, 36, 39\}$

$A \cap B = \{28, 31\}$

$A \oplus B = \{21, 23, 24, 25, 26, 33, 34, 36, 39\}$ = 9 anggota



Dapat diambil kesimpulan bahwa pelanggan yang berasal dari 3 sumber yaitu 28, 31

Paramoter Porsonalisas:

$N = 21 ; a = 2 ; b = 1 ; k = 4 ; 100 = 650 ; \alpha = 4100$

$f(x) = K \cdot x + (a+b) = 4x + (2+1) = 4x + 3$

$g(x) = \frac{x^2}{(b+1)} + a = \frac{x^2}{(1+1)} + 2 = \frac{x^2}{2} + 2$

$(f \circ g)(x) = f(g(x)) = 4\left(\frac{x^2}{2} + 2\right) + 3 = 4 \cdot \frac{x^2}{2} + 4 \cdot 2 + 3$

$= 2x^2 + 8 + 3 = 2x^2 + 11$

$(g \circ f)(x) = g(f(x)) = \frac{(4x+3)^2}{2} + 2 = \frac{(16x^2 + 24x + 9)}{2} + 2$

$= \frac{16x^2 + 24x + 9}{2} + \frac{4}{2} = \frac{16x^2 + 24x + 13}{2}$

$(f \circ g)(3) = 2(3^2) + 11 = 2(9) + 11 = 18 + 11 = 29$

$(g \circ f)(3) = \frac{16(3^2) + 24(3) + 13}{2} = \frac{16(9) + 72 + 13}{2}$

$= \frac{144 + 85}{2} = \frac{229}{2}$

Komutatif apabila $g \circ f = f \circ g$, karena $x = b + 2 = 1 + 2 = 3$ maka $f \circ g(x) = 29$, $g \circ f(x) = \frac{229}{2}$ Kesimpulan tidak memenuhi syarat.

D) $f(x) = 4x + 3$

$f^{-1}(x) = y = 4x + 3 \rightarrow x = \frac{y-3}{4} \rightarrow f^{-1}(x) = \frac{x-3}{4}$

$x = (K \cdot 100 + a + b) = (4 \cdot 100 + 2 + 1)$ 403 ribu

$f^{-1}(x) = \frac{403 - 3}{4} = \frac{400}{4} = 100$

E) $f(x) = 4x + 3 \rightarrow$ Bijektif karena injektif dan surjektif

* Injektif (satu-satu) $f(a) = f(b) \Rightarrow a = b$ $4a + 3 = 4b + 3$ $4a = 4b$ $a = b$ f = Injektif, karena $a = b$. yang berarti tidak ahan x yang berbeda	* Surjektif (pada) $y = 4x + 3$ $\frac{y-3}{4} = x$ $y = 3 = \frac{3-3}{4} = \frac{0}{4} = 0$ $f(x)$ = ~~injek~~ surjektif, karena hasil nya selalu ada. mau berapun y nya dan merupakan bil real bukan bil imajiner

No
Date

$g(x) = \frac{x^2}{2} + 2 \Rightarrow$ bukan bijektif karena

$\frac{a^2}{2} + 2 = \frac{b^2}{2} + 2$ injektif X

surjektif X

$\frac{a^2}{2} = \frac{b^2}{2}$

$a^2 = b^2$

$a = \pm b$

Bukan injektif karena $a = \pm b$, yang menyebabkan ada dua x yang berbeda untuk menghasilkan hasil yg sama

$y = \frac{x^2}{2} + 2$

$2(y-2) = x^2$

$y = 1 \Rightarrow x^2 = 2(1-2)$

$x^2 = 2(-1)$

$x^2 = -2$

$x = \sqrt{-2}$

$y = \geq 2 \; (2, \infty)$, menyebabkan bukan surjektif karena y nya tidak bisa semaunya dan ada kalanya hasil x nya bilimajiner